

Algorithms Problem Solving

Level 1

Triangle Area

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Definition

- ◆ This program reads the base and height of a triangle and calculates its area.
- ◆ The program uses functions, reference parameters, multiplication, division, and float values.

Explanation

- ◆ The `readNumbers()` function reads the base `B` and height `H` of the triangle.
- ◆ The parameters `int& B` and `int& H` are reference parameters, so the function stores the input directly in the variables from `main()`.
- ◆ The `calculateArea()` function calculates the triangle area using the base and height.
- ◆ The formula is:
$$\text{Area} = (B \times H) / 2$$
- ◆ The function returns the calculated area as a float.
- ◆ The `printTriangleArea()` function displays the base, height, and calculated area.
- ◆ It calls `calculateArea(B, H)` to get the result.

Steps in main()

1. Create two integer variables for the base and height.
2. Read the base and height.
3. Pass the values to calculateArea().
4. Multiply the base by the height.
5. Divide the result by 2.
6. Return the calculated area.
7. Print the triangle area.

💡 Example

◆ If the user enters:

Base: 10

Height: 6

◆ The area is calculated as:

$$(10 \times 6) / 2 = 30$$

➡ Result

The area of the triangle with base 10 and height 6 is: 30

Note

- ◆ The formula for the area of a triangle is half the area of a rectangle with the same base and height.
- ◆ The function uses float parameters and return type so the calculation can produce decimal values.

⊘ Warning

⚡ The program does not validate the input. Negative or zero values are accepted even though they do not represent a normal triangle's base and height.



Comparison

Rectangle

- ✓ Area = length $\times$ width.
- ✓ Uses the full product of the two dimensions.
- ✓ Does not divide the result by 2.

Triangle

- ✓ Area = (base $\times$ height) / 2.
- ✓ Uses the same multiplication idea.
- ✓ Divides the result by 2 because the triangle represents half of a corresponding rectangle.

✦✦ Summary

- ✓ The area of a triangle is calculated using $\text{base} \times \text{height} \div 2$.
- ✓ Reference parameters allow `readNumbers()` to modify variables from `main()`.
- ✓ A function can return a calculated float value.
- ✓ Separating input, calculation, and output makes the program easier to understand.

Thank you 😊